



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

**NAAC
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Major Project/Internship

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**NSE Stock AI: Real-Time Buy/Sell
Prediction System**

Name :Sojitra Jaimin R.

Enrollment No:2201031000127

Branch: IT

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Introduction

- Stock market prediction uses Artificial Intelligence and Machine Learning.
- Traditional trading depends on human judgment and emotion.
- NSE Stock AI predicts short-term stock movement using historical and live data.
- The system gives BUY / SELL output with confidence

Background and Motivation

- Financial markets generate large time-series data.
- Manual analysis of stock data is complex and time-consuming.
- Machine learning helps identify hidden market patterns.
- Motivation is to build an automated intelligent prediction system.

Relevance and Importance

- Helps investors make informed decisions.
- Reduces emotional bias in trading.
- Demonstrates real-world AI application in finance.
- Supports financial analytics and risk management

Literature review

- Traditional models like ARIMA are used for time-series forecasting
- Machine learning models improve prediction performance.
- Technical indicators like SMA, RSI, and MACD are widely used.
- Ensemble models often perform better on financial data.

objectives

- Collect historical stock market data.
- Perform exploratory data analysis.
- Generate technical indicators using feature engineering.
- Develop machine learning prediction models.
- Evaluate model performance.
- Visualize prediction results

Required Tools and Technology

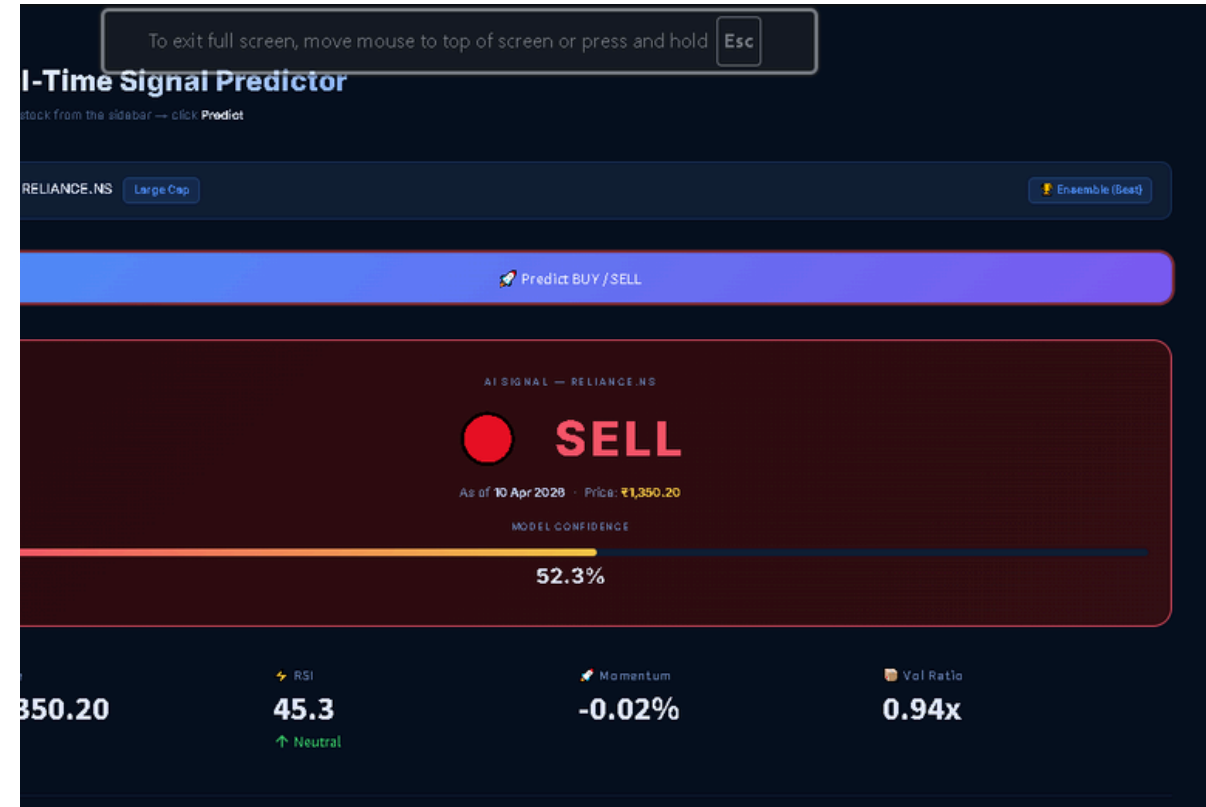
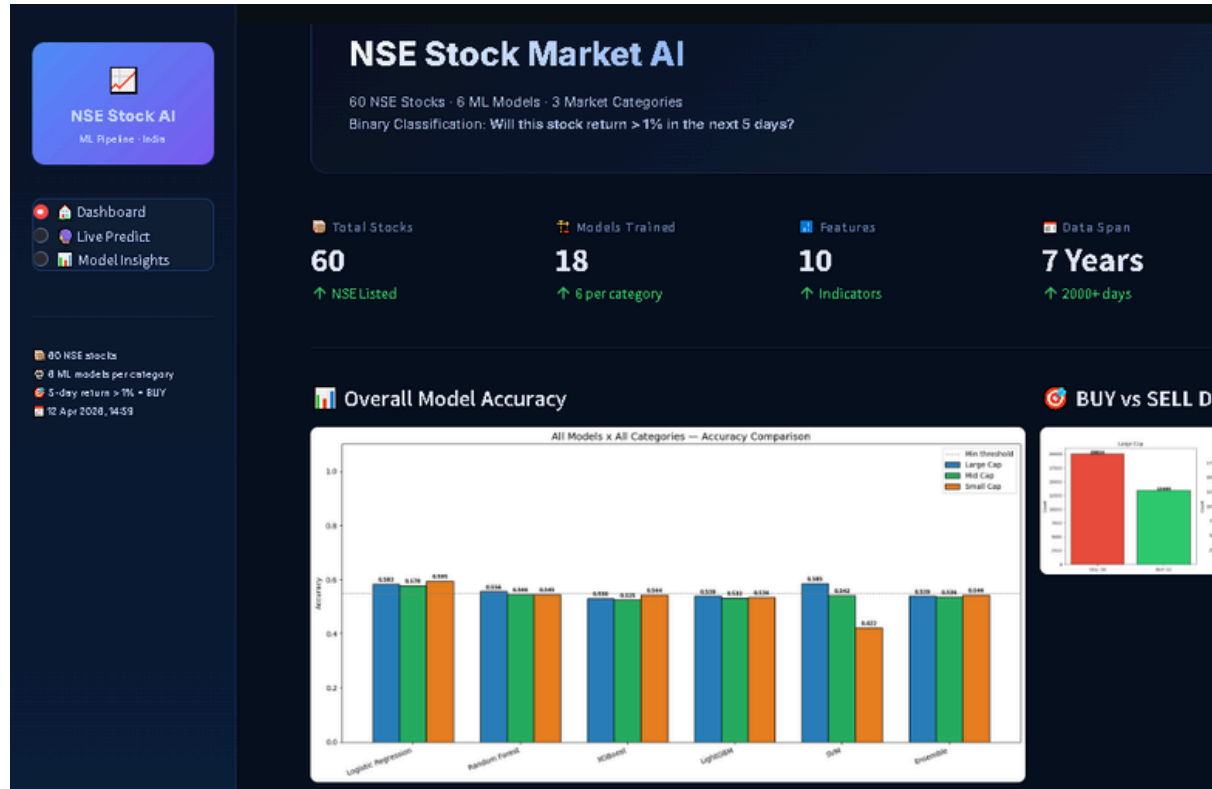
Programming & Libraries :

- Programming Language: Python
- Libraries: Pandas, NumPy, Plotly, Matplotlib
- Machine Learning: Scikit-learn, XGBoost, LightGBM
- Data Source: Yahoo Finance (yfinance)
- Technical Indicators: TA library
- Environment: Jupyter Notebook / VS Code / Streamlit

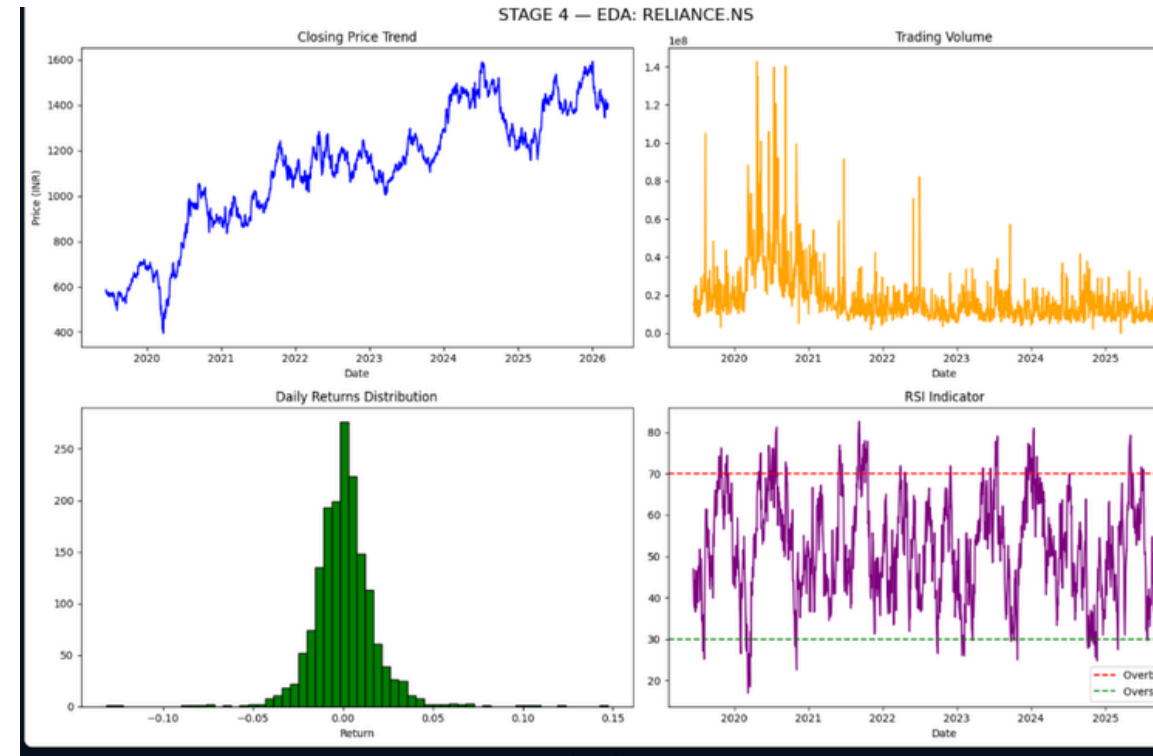
Method/ Approach

- Collect stock data using yfinance API.
- Perform data preprocessing and cleaning.
- Apply feature engineering using SMA, RSI, MACD, EMA, ATR, Momentum, Volume Ratio, and Bollinger Band Width.
- Train machine learning models on the generated features.
- Predict BUY / SELL signal for the latest data row.
- Display confidence and charts in the dashboard

Output / Conclusion



Output / Conclusion



Output / Conclusion

- The system automates stock analysis using technical indicators and machine learning models.
- It reduces human bias and improves trading decisions through data-driven insights.
- The integration of multiple models enhances prediction accuracy and reliability.
- The Streamlit dashboard provides a simple and interactive user experience.
- The project demonstrates the practical application of AI in financial analytics.

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